CASE RECORDS of the MASSACHUSETTS GENERAL HOSPITAL

Case 34-2023: A 49-Year-Old Woman with Loss of Consciousness and Thrombocytopenia

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PRESENTATION OF CASE

Dr. Joseph B. Stegeman (Emergency Medicine): A 49-year-old woman was evaluated in the emergency department of this hospital after being found unconscious at work.

At approximately 4:30 p.m. on the day of this evaluation, the patient was found unconscious on the floor next to a copy machine at work. Emergency medical services were activated and found the patient to be confused and aphasic; she could say only “yes” and “no.” There was no evidence of incontinence or trauma. A fingerstick blood glucose level was normal. The patient was transported to the emergency department of this hospital for evaluation.

On the patient’s arrival at the emergency department, limited additional information was obtained from coworkers and family members. The patient had had coronavirus disease 2019 (Covid-19) 4 weeks earlier and had returned to work after 2 weeks. On the morning before this evaluation, coworkers had observed that she appeared “unwell.” Before 2 p.m., she had participated in a lunchtime conference call and had spoken to family members and colleagues. Her medical history was notable for hypertension, for which she took amlodipine. There was no known family history of neurologic disorders.

On examination, the temporal temperature was 36.3°C, the heart rate 70 beats per minute, the blood pressure 141/98 mm Hg, the respiratory rate 18 breaths per minute, and the oxygen saturation 96% while the patient was breathing ambient air. On examination, she was aphasic; her speech was described as mumbling, and she was able to answer simple questions only intermittently. Strength and sensation appeared to be grossly intact, but she could not perform rapidly alternating movements or tests of coordination. There was no evidence of head or other trauma. The remainder of the examination was normal. Imaging studies were obtained.

Dr. Shahmir Kamalian: Forty minutes after the patient’s arrival, computed tomography (CT) angiography of the head and neck (Fig. 1), performed after the adminis-

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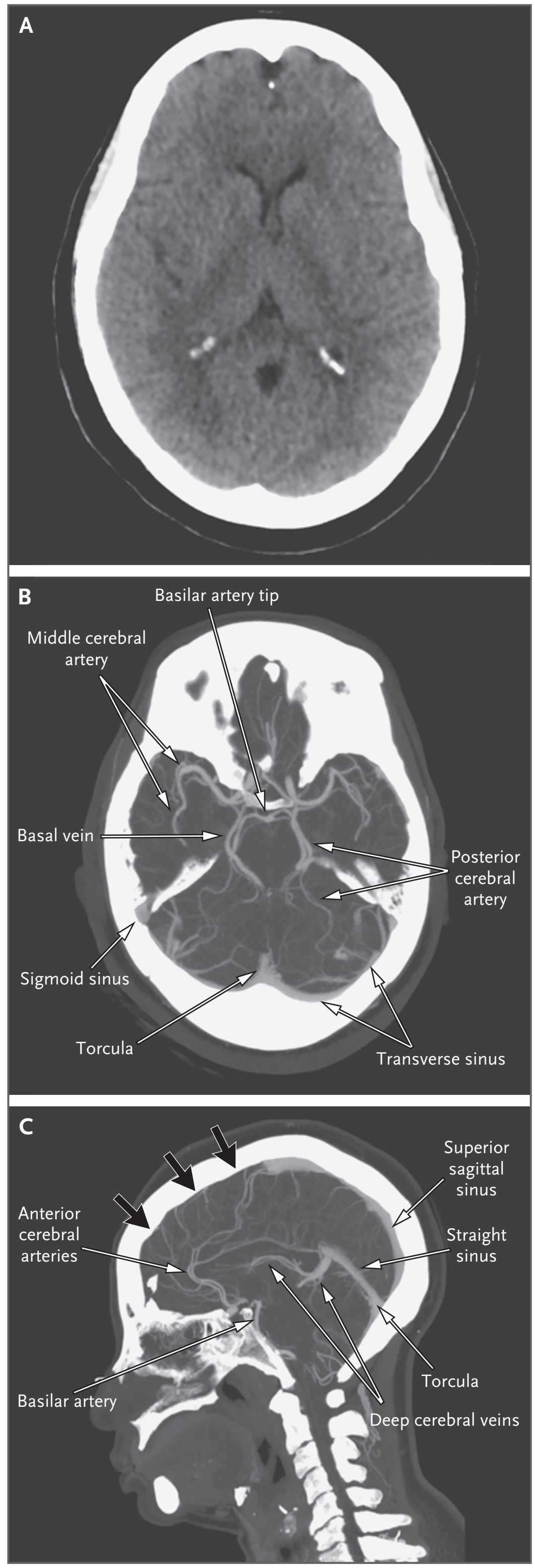

Figure 1. CT and CT Angiography of the Head.
An axial image from CT of the head (Panel A), performed after the administration of intravenous contrast material, shows preserved gray matter–white matter differentiation, with no evidence of hemorrhage or mass effect. Axial and sagittal maximum-intensity-projection images from CT angiography of the head (Panels B and C, respectively) show that the major intracranial arteries are all unremarkable, with no stenosis, irregularities, or occlusion. The rostrum of the superior sagittal sinus is hypoplastic (black arrows), which is a normal variant. Otherwise, the dural venous sinuses and deep cerebral veins are patent.

tration of intravenous contrast material, showed no evidence of infarction, hemorrhage, or intracranial or cervical arterial stenosis or occlusion. The dural venous sinuses and deep cerebral veins were patent. A chest radiograph was normal.

Dr. Stegeman: Approximately 1 hour after the patient's arrival, initial laboratory test results were received (Table 1). The blood ethanol level was undetectable, and urine toxicologic screening was negative, as was screening for human chorionic gonadotropin. The blood thyrotropin level was normal. Intravenous normal saline was administered.

Within 2 hours after the patient's arrival, her aphasia abated; her speech was slow but fluent. On a repeat review of systems, she noted that she had felt warm earlier in the day, but she reported no other symptoms and no use of alcohol or drugs.

Three hours after the patient's arrival, additional laboratory test results were received (Table 1). Analysis of a peripheral-blood smear showed numerous schistocytes per high-power field, very few platelets, and reticulocytosis.

Diagnostic tests were performed, and management decisions were made.

DIFFERENTIAL DIAGNOSIS

Dr. Kori S. Zachrison: This 49-year-old woman with hypertension was found unconscious next to a copy machine after coworkers had observed that she looked "unwell" earlier in the day. An examination in the emergency department was notable for aphasia and difficulty with coordination testing. From the perspective of the evaluating

Table 1. Laboratory Data.*

Variable	Reference Range, Adults†	1 Hr after Arrival	3 Hr after Arrival
Hemoglobin (g/dl)	12.0–16.0	8.5	7.6
Hematocrit (%)	36.0–46.0	25.4	22.1
White-cell count (per μl)	4500–11,000	16,200	19,570
Platelet count (per μl)	150,000–400,000	5000	7000
High-sensitivity troponin T (ng/liter)	0–9	72	105
Erythrocyte sedimentation rate (mm/hr)	0–20	50	—
Lactate (mmol/liter)	0.5–2.0	2.4	—
C-reactive protein (mg/liter)	<8.0	26.7	—
Sodium (mmol/liter)	135–145	138	—
Potassium (mmol/liter)	3.4–5.0	2.5	—
Chloride (mmol/liter)	98–108	97	—
Carbon dioxide (mmol/liter)	23–32	27	—
Urea nitrogen (mg/dl)	8–25	21	—
Creatinine (mg/dl)	0.60–1.50	1.07	—
Glucose (mg/dl)	70–110	130	—
Alanine aminotransferase (U/liter)	7–33	18	—
Aspartate aminotransferase (U/liter)	9–32	50	—
Alkaline phosphatase (U/liter)	45–115	50	—
Total bilirubin (mg/dl)	0.0–1.0	2.6	—
Direct bilirubin (mg/dl)	0.0–0.4	0.3	—
Prothrombin time (sec)	11.5–14.5	14.8	15.2
International normalized ratio	0.9–1.1	1.2	1.2
Partial-thromboplastin time (sec)	22.0–36.0	—	28.8
D-dimer (ng/ml)	<500	—	3857
Fibrinogen (mg/dl)	150–400	—	291
Haptoglobin (mg/dl)	30–200	—	<10
Lactate dehydrogenase (U/liter)	110–210	—	1495
Reticulocytes (%)	0.7–2.5	—	5.8

* To convert the values for lactate to milligrams per deciliter, divide by 0.1110. To convert the values for urea nitrogen to millimoles per liter, multiply by 0.357. To convert the values for creatinine to micromoles per liter, multiply by 88.4. To convert the values for glucose to millimoles per liter, multiply by 0.05551. To convert the values for bilirubin to micromoles per liter, multiply by 17.1.

† Reference values are affected by many variables, including the patient population and the laboratory methods used. The ranges used at Massachusetts General Hospital are for adults who are not pregnant and do not have medical conditions that could affect the results. They may therefore not be appropriate for all patients.

emergency physician, there are two equally important questions to consider: why did the patient fall and lose consciousness, and did any injuries result from this event?

FALLS AND LOSS OF CONSCIOUSNESS

A person may fall or lose consciousness for many reasons. Neurologic events — such as stroke, seizure, or another neurologic condition — and syncopal events are common causes. Other considerations include a toxic or metabolic cause, a mechanical fall, or a functional neurologic disorder or conversion disorder. This patient had focal neurologic findings, specifically aphasia and apraxia, and she had a last known well time of less than 3 hours. Therefore, stroke needs to be ruled out first, with rapid neuroimaging and evaluation.

Aphasia is suggestive of a classic left middle cerebral artery (MCA) stroke, whereas coordination difficulties typically localize to the cerebellum, although they might be attributed to weakness or visuospatial abnormalities in this patient. An alternative explanation for aphasia and cerebellar findings may be multifocal infarcts, such as those seen with cardioembolism, Covid-19–associated vasculitis, or reversible cerebral vasoconstriction syndrome. In addition, patients with cerebral venous thrombosis can present with aphasia and focal neurologic signs that do not fit a specific vascular distribution.

It is possible that the patient had an event localized to the cerebellum, rather than a multifocal process. Cerebellar aphasia has been described,[1] but it is much less common than left MCA stroke. Acute disseminated encephalomyelitis and multiple sclerosis are both uncommon causes of aphasia; acute disseminated encephalomyelitis has been described in association with Covid-19.[2] Because the patient had altered mental status and had looked unwell earlier in the day, other considerations include infectious, structural, or encephalopathic processes, such as cerebral or cerebellar abscess, cerebral or cerebellar neoplasm, encephalitis, meningitis, or Hashimoto's encephalopathy.

It is notable that the patient's aphasia abated without specific therapeutic intervention. Spontaneous resolution of such neurologic deficits is suggestive of transient ischemic attack or seizure. Migraine with brain-stem aura is another possibility, although it is very rare.[3] Syncopal events — whether vasovagal, orthostatic, or cardiogenic (caused by structural heart disease or arrhythmia) — would be unlikely to result in focal neurologic findings, but it is possible that the patient had focal concussive symptoms that were decreasing over time.

A toxic cause related to the copy machine is unlikely because the patient had appeared unwell before she had used it. Exposure to an environmental toxin, such as carbon monoxide, is also unlikely, given that other people in the office were not affected.

RESULTING INJURIES

As we consider the cause of the patient's fall and loss of consciousness, we must also consider what injuries, if any, resulted from the event. Although she presumably fell from standing, traumatic injuries from the fall must still be considered, such as intracerebral hemorrhage, cervical spine injury, or a vertebral- or carotid-artery dissection that led to the current focal neurologic findings. An evaluation for thoracoabdominal trauma and for bony or skin injury from the fall is also warranted.

LABORATORY AND IMAGING FINDINGS

While the patient was in the emergency department, some key laboratory test results became available. She had a negative test for human chorionic gonadotropin, a finding that makes a diagnosis of seizure due to eclampsia unlikely (unless she were recently postpartum, which she was not). She had a normal thyrotropin level, which rules out Hashimoto's encephalopathy, and an undetectable ethanol level, which rules out acute alcohol intoxication.

The results of CT angiography rule out vertebral- or carotid-artery dissection, subarachnoid hemorrhage, intracranial hemorrhage or mass, and cerebral venous thrombosis. The absence of vascular beading on CT angiography rules out reversible cerebral vasoconstriction syndrome. Ischemic stroke and transient ischemic attack cannot be ruled out on the basis of normal findings on CT, although a large-vessel occlusion would be expected with these conditions.

It is notable that the patient's potassium level was 2.5 mmol per liter. Hypokalemia suggests an arrhythmogenic cause of syncope, perhaps related to a prolonged QT interval, torsades de pointes, or a run of ventricular tachycardia. Telemetry and electrocardiography are indicated in this case.

The patient had anemia and thrombocytopenia, findings that increase the concern for thoracoabdominal injury with internal bleeding resulting from the trauma of her fall. Further diagnostic evaluation with either an extended bedside ultrasonographic examination (known as focused assessment with sonography for trauma, or FAST) or CT of the chest, abdomen, and pelvis may be performed.

In the presence of neurologic findings, anemia and thrombocytopenia together also suggest the possibility of a thrombotic microangiopathy, such as thrombotic thrombocytopenic purpura (TTP). This constellation of findings might lead us to consider systemic lupus erythematosus with a thrombotic microangiopathy or catastrophic antiphospholipid syndrome (CAPS).

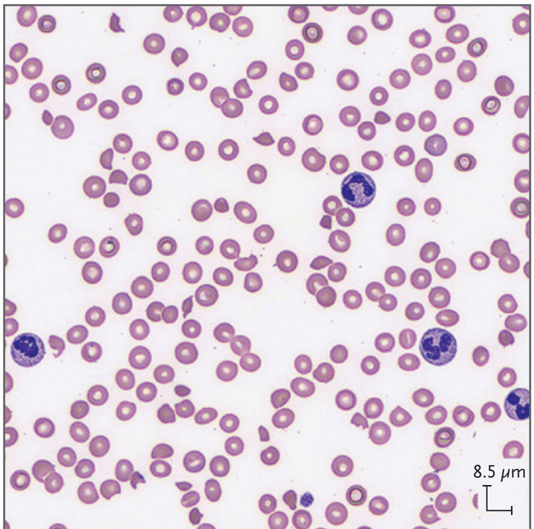

Figure 2. Peripheral-Blood Smear.
Wright–Giemsa staining shows evidence of multiple schistocytes, as well as very few platelets.

EMERGENCY DEPARTMENT COURSE

The improvement in the patient's condition and resolution of aphasia subsequently enabled a full review of systems, which was notable only for a subjective warm temperature, with no other symptoms reported. At this point, encephalitis and meningitis could be ruled out on the basis of her improved mental status and the absence of suggestive symptoms. On additional laboratory testing, there was a further decrease in the hemoglobin level, a persistent reduction in the platelet count, a normal international normalized ratio, and elevated D-dimer and lactate dehydrogenase levels. The detection of a normal fibrinogen level, a low haptoglobin level, and reticulocytes and schistocytes on a peripheral-blood smear makes TTP the most likely unifying diagnosis in this patient. TTP can be idiopathic, precipitated by an identifiable trigger such as an infection, or related to an underlying autoimmune disorder, CAPS, or cancer. It is unlikely that the patient had drug-related TTP, given that amlodopine was her only medication.

CLINICAL IMPRESSION AND INITIAL MANAGEMENT

Dr. David B. Sykes: In this patient with pronounced anemia and thrombocytopenia as well as changes in mental status, certain laboratory test results helped to narrow the differential diagnosis. The elevated lactate dehydrogenase level, elevated indirect bilirubin level, and low haptoglobin level suggested that the anemia had resulted from hemolysis. The normal prothrombin time, international normalized ratio, and activated partial-thromboplastin time argued strongly against a consumptive process, such as disseminated intravascular coagulation. The elevated reticulocyte count, although inappropriately low given the degree of anemia, essentially ruled out aplastic anemia and vitamin B_{12} deficiency.

Microscopic evaluation of a peripheral-blood smear (Fig. 2) confirmed the degree of thrombocytopenia and was particularly striking for the increased number of schistocytes. The presence of schistocytes distinguishes microangiopathic hemolytic anemia from other causes of anemia (i.e., autoimmune hemolytic anemia or aplastic anemia) and from other causes of thrombocytopenia (e.g., immune thrombocytopenic purpura or heparin-induced thrombocytopenia). Quantification of schistocytes can be challenging. The presence of 1 or 2 schistocytes per high-power field is often a nonspecific finding in patients with acute illness, especially those with sepsis or those receiving mechanical support (e.g., extracorporeal membrane oxygenation, hemodialysis, continuous venovenous hemofiltration, or treatment with ventricular assist devices). However, this patient's peripheral-blood smear showed 5 or 6 schistocytes per high-power field, which accounted for approximately 5% of the red cells. This striking number of schistocytes was consistent with the degree of anemia and thrombocytopenia and confirmed the presence of microangiopathic hemolytic anemia.

In a patient with changes in mental status, the presence of microangiopathic hemolytic anemia is highly suggestive of a diagnosis of acquired TTP. Acquired TTP is an autoimmune condition driven by antibody-mediated clearance of the plasma enzyme ADAMTS13 (a disintegrin and metalloproteinase with thrombospondin motif 13). Confirmatory laboratory testing for ADAMTS13 usually takes 1 to 3 days. Therefore, empirical therapy is often initiated while test results are pending. Transfusion of fresh-frozen plasma is associated with a low risk of harm and has the potential benefit of providing replacement of ADAMTS13 until the initiation of definitive therapy.

CLINICAL DIAGNOSIS

Thrombotic thrombocytopenic purpura.

DR. KORI S. ZACHRISON'S DIAGNOSIS

Thrombotic thrombocytopenic purpura.

DIAGNOSTIC TESTING

Dr. Patricia A.R. Brunker: In determining the cause of thrombotic microangiopathy, testing for ADAMTS13 activity and reflex testing for the ADAMTS13 inhibitor are essential. ADAMTS13 is a plasma protease synthesized in the liver that cleaves von Willebrand factor at a site buried in the core β-sheet of the A2 domain. Most cases of TTP are caused by an autoantibody to ADAMTS13 that prevents this cleavage. When the cleavage does not occur, platelet microthrombi form in small vessels in multiple anatomical sites, causing the clinical manifestations of TTP.

ADAMTS13 activity is measured by means of fluorescence resonance energy transfer. Cleavage of a synthetic substrate separates a quencher molecule from a fluorescent tag, and the emitted fluorescence (reported as a percentage of normal function) is proportional to enzyme activity. Although a mild decrease in ADAMTS13 activity can be seen with aging and certain medical conditions (e.g., liver disease, inflammatory conditions, or pregnancy), a severe decrease in ADAMTS13 activity (to a level of <10%) is characteristic of TTP. In less than 5% of cases, such ADAMTS13 deficiency is genetic, causing congenital TTP. To distinguish congenital TTP (also known as the Upshaw–Schulman syndrome) from acquired TTP, an inhibitor assay is performed as a reflex test when the level of ADAMTS13 activity is low. As with other coagulation inhibitor assays, one inhibitor unit is defined as the level at which the inhibitor, in an equal volume of normal plasma, reduces enzyme activity by one half. In this patient, the ADAMTS13 activity level was severely reduced (<5%; reference value, >67%), and the inhibitor was present (1.4 inhibitor units; reference value, ≤0.4).

LABORATORY DIAGNOSIS

Acquired thrombotic thrombocytopenic purpura.

DISCUSSION OF MANAGEMENT

Dr. Brunker: The ADAMTS13 panel is typically performed at a regional reference laboratory, and results may not be available for several days. However, TTP is a life-threatening emergency that requires immediate treatment, so clinical diagnostic tools may be used to inform decision making. The decision to initiate the best available therapy — therapeutic plasma exchange — often must be made before the results of diagnostic laboratory testing have been received.

Therapeutic plasma exchange is an invasive procedure that involves the use of apheresis services, and it is not readily available in many care settings. The patient may need to be transferred to a medical center where the appropriate staff and equipment are available. Therapeutic plasma exchange is associated with procedural risks and considerable use of health care resources. Plasmapheresis for the treatment of TTP exposes the patient to a high volume of donor plasma, and central venous access needs to be obtained with the placement of a rigid catheter that is compatible with the high flow of apheresis instruments. In deciding whether therapeutic plasma exchange should be started immediately, before the results of ADAMTS13 testing are available, the PLASMIC score is particularly instructive.[4]

Developed in 2017, the PLASMIC score is based on seven components: a platelet count (P) of less than 30,000 per microliter; the presence of hemolysis (L) as indicated by the laboratory value for indirect bilirubin, reticulocytes, or haptoglobin; the presence of active cancer (A) in the past 12 months, in the absence of stem-cell or solid-organ transplantation (S); a mean corpuscular volume (M) of less than 90 fl; an international normalized ratio (I) of less than 1.5; and a creatinine level (C) of less than 2.0 mg per deciliter (177 mmol per liter). All these components can be assessed with the initial history taking, physical examination, and routine laboratory testing. Therapeutic plasma exchange is typically considered when the patient has at least five components and thus the PLASMIC score is 5 or greater, indicating at least intermediate risk of TTP.

To determine whether emergency therapeutic plasma exchange or plasma transfusion is appropriate, a hematology consultation is obtained, along with a transfusion medicine consultation. Because TTP is a rare condition, the transfusion medicine specialist may be the on-call consultant with the most experience with TTP and therefore can offer important diagnostic and therapeutic services to the patient.

THERAPEUTIC PLASMA EXCHANGE

Therapeutic plasma exchange not only permits the transfusion of a high volume of ADAMTS13 in donor plasma but also removes the circulating antibody inhibitor in more than 95% of patients with autoimmune TTP. Plasma transfusion may be started before the initiation of therapeutic plasma exchange as a temporizing measure. A widely accepted management strategy is the administration of therapeutic plasma exchange in daily cycles, involving the exchange of 1.0 plasma volumes until the platelet count reaches at least 150,000 per microliter, for 3 consecutive days. In patients with severe disease, the exchange of 1.5 plasma volumes may be necessary.

This patient underwent six cycles of therapeutic plasma exchange, with the exchange of 1.0 plasma volumes, during the first 5 hospital days, and the platelet count increased to 396,000 per microliter. At many institutions, the administration of therapeutic plasma exchange is tapered from daily to every other day, which allows the clinician to determine whether the platelet count has stabilized before the central catheter is removed. We recommend keeping the central catheter in place for a few days after the last cycle of therapeutic plasma exchange, while monitoring the platelet count, so that treatment can be resumed if necessary. This patient was discharged on hospital day 8 with a tunneled catheter in place for this purpose.

ADDITIONAL CONSIDERATIONS

For both immediate and postdischarge treatment of TTP, therapeutic plasma exchange must be accompanied by immunosuppression. Glucocorticoid therapy is typically started alongside therapeutic plasma exchange[5] and was started in this patient on hospital day 3.

The administration of rituximab within 3 days after presentation is associated with better clinical outcomes — specifically, fewer days of treatment with therapeutic plasma exchange, fewer hospital days overall, a higher ADAMTS13 activity level, a lower inhibitor level, and a lower risk of relapse (10%, vs. >50% without rituximab).[6,7] This patient received her first rituximab infusion on hospital day 5.

Once the platelet count reaches at least 50,000 per microliter, treatment with low-dose aspirin may be helpful.[8] Treatment with several other agents, including bortezomib and daratumumab, has been described in case reports and case series, typically in patients with TTP that is refractory to rituximab.[9]

Dr. Sykes: Although causation is challenging to prove, TTP seems to develop frequently in the context of either an immune trigger, such as a viral infection, or immune dysregulation related to another autoimmune disease or active cancer. In this patient, Covid-19 had occurred 4 weeks before the diagnosis of acquired TTP and seems to be the most likely trigger. However, rectal cancer was diagnosed in this patient 2 months after the diagnosis of acquired TTP. Whether these conditions are directly related is unclear.

After the diagnosis of TTP, patients receive counseling regarding relapse. The lifetime risk of relapse is at least 20%, even in patients who receive rituximab therapy. In our practice, we tend to monitor laboratory test results, including the complete blood count and lactate dehydrogenase level, on a weekly basis for 1 month. Thereafter, monitoring is performed every 2 weeks for 2 months and then performed monthly for the first year after diagnosis. Anecdotally, we find that many patients are able to "sense" a relapse even before it is overtly evident in their laboratory test results, so patient symptoms are also monitored.

Repeat testing for ADAMTS13 can be performed while the patient is in clinical remission. During remission, the ADAMTS13 inhibitor clears and the activity level normalizes in many cases, but not all. Some patients can be clinically well and have normal laboratory test results but have a low ADAMTS13 activity level. There is no clear recommendation regarding treatment for these patients, but they seem to be at higher risk for relapse.

In very rare cases, the ADAMTS13 activity level remains very low (<10%) during remission.

In these cases, an evaluation for congenital TTP can be performed with sequencing of *ADAMTS13*.[10]

FOLLOW-UP

Dr. Sykes: This patient completed four weekly infusions of rituximab and a 2-month tapering course of glucocorticoids. Approximately 6 months after the diagnosis of acquired TTP, the patient had a relapse, for which she received therapeutic plasma exchange and glucocorticoid therapy. It is possible that her underlying cancer was partially responsible for the relapse. She also received a second course of rituximab to reduce the chance of recurrence.

A Physician: When should platelet transfusion be considered for a patient with TTP?

Dr. Sykes: Although I hesitate to make an overgeneralizing statement, platelet transfusions should not be given to patients who are not bleeding. I would recommend avoiding prophylactic platelet transfusion, regardless of the platelet count. When profound thrombocytopenia (a platelet count of <10,000 per microliter) occurs in patients who are not actively bleeding, prophylactic platelet transfusions are often either unhelpful (in the case of immune thrombocytopenic purpura) or possibly harmful (in the case of heparin-induced thrombocytopenia, TTP, or disseminated intravascular coagulation).

FINAL DIAGNOSIS

Acquired thrombotic thrombocytopenic purpura.

This case was presented at Emergency Medicine Grand Rounds.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

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